

Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030071

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Query 1: Create the database

```
1 2520030071-G. Hemanth Venkatesh
2 CREATE DATABASE IF NOT EXISTS bookflow5_db;
```

Automatic context disabled. Use the manually get help for current caret position toggle automa

Output: -

✓	2	10:09:03	CREATE DATABASE IF NOT EXISTS bookflow5_db	1 row(s) affected	0.094 sec
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Query 2: Select the database

```
SELECT DATABASE() AS current_database;
2520030071-G. Hemanth Venkatesh
USE bookflow5_db;
```

Output: -

✓	3	10:11:29	USE bookflow5_db	0 row(s) affected	0.000 sec
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Query 3: Create the books table

```

1  2520030071-G. Hemanth Venkatesh
2  CREATE TABLE books (
3  book_id INT AUTO_INCREMENT PRIMARY KEY,
4  title VARCHAR(255) NOT NULL,
5  isbn VARCHAR(13) NOT NULL UNIQUE,
6  published_year INT,
7  CONSTRAINT chk_published_year CHECK (published_year<2027)
   );

```

Output: -

✓	4	10:18:46	CREATE TABLE books (book_id INT AUTO_INCR...	0 row(s) affected	0.125 sec
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Query 4: Create the members table

```

1  2520030071-G. Hemanth Venkatesh
2  CREATE TABLE members (
3  member_id INT AUTO_INCREMENT PRIMARY KEY,
4  full_name VARCHAR(100) NOT NULL,
5  email VARCHAR(150) NOT NULL UNIQUE
6  );

```

Output: -

✓	5	10:23:27	CREATE TABLE members (member_id INT AUTO_I...	0 row(s) affected	0.079 sec
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Query 5 (verification): Inspect the table structures

```

1  2520030071-G. Hemanth Venkatesh
2
3  DESCRIBE books;

```

Output: -

✓	6	10:26:25	DESCRIBE books	4 row(s) returned	0.047 sec / 0.000 sec
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Result Grid


Filter Rows:

Export:


Wrap Cell Content:


	Field	Type	Null	Key	Default	Extra
▶	book_id	int	NO	PRI	NULL	auto_increment
	title	varchar(255)	NO		NULL	
	isbn	varchar(13)	NO	UNI	NULL	
	published_year	int	YES		NULL	


Form Editor


Field Types

Query 6: Insert 3 books

```

1  2520030071-G. Hemanth Venkatesh
2  INSERT INTO books (title,isbn,published_year) VALUES
3  ('The Alchemist','9780061122415',1988),
4  ('The autobiography','9780132350884',2005),
5  ('Harry Potter','9780735211292',1997);

```

[Context Help](#)

Output: -

✓	8	10:33:11	INSERT INTO books (title,isbn,published_year) VAL...	3 row(s) affected	Records: 3	Duplicates: 0	Warnings: 0	0.031 sec
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Query 7: Insert 3 members

```

1  2520030071-G. Hemanth Venkatesh
2  INSERT INTO members (full_name,email) VALUES
3  'G. Hemanth Venkatesh','hemanthvenkatesh5554@gmail.com'
4  ('Aditya Rao','aditya.rao@gmail.com'),
5  ('Neha Sharma','neha.sharma@gmail.com');

```

[Context Help](#)

Output: -

✓	9	10:38:04	INSERT INTO members (full_name,email) VALUES (...)	3 row(s) affected	Records: 3	Duplicates: 0	Warnings: 0	0.047 sec
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Query 8: Display

```

1  SELECT * FROM books;

```


Output: -

✓	10	10:40:26	SELECT * FROM books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
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Result Grid

Filter Rows:

Edit:

Export/Import:

SELECT * FROM members;

✓	11	10:41:15	SELECT * FROM members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
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Result Grid

 Filter Rows:

Edit: 

Export/Import: 

	member_id	full_name	email
▶	1	G. Hemanth Venkatesh	hemanthvenkatesh5554@gmail.com
	2	Aditya Rao	aditya.rao@gmail.com
	3	Neha Sharma	neha.sharma@gmail.com
✱	NULL	NULL	NULL

Result Grid

 Form Editor

Query 9: Constraint Tests (These MUST Fail)

Test 1: Duplicate ISBN (UNIQUE violation)

```
1 2520030071-G. Hemanth Venkatesh
2 INSERT INTO books (title, isbn, published_year)
  VALUES ('Fake Copy', '9780061122415', 2000);
```

Output: -

✗	12	10:47:42	INSERT INTO books (title, isbn, published_year) VALUES ...	Error Code: 1062. Duplicate entry '9780061122415' for key...	0.047 sec
---	----	----------	--	--	-----------

Test 2: NULL title (NOT NULL violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES (NULL, '9999999999999', 2010);
```

Output: -

✖	14	10:50:07	INSERT INTO books (title, isbn, published_year) VALUES (NULL, '9999999999999', 2010)	Error Code: 1048. Column 'title' cannot be null	0.000 sec
---	----	----------	--	---	-----------

Test 3: Future publication year (CHECK violation)

```
INSERT INTO books (title, isbn, published_year)
VALUES ('The Alchemist', '8888888888888', 2030);
```

Output: -

✖	15	10:52:21	INSERT INTO books (title, isbn, published_year) VALUES ('The Alchemist', '8888888888888', 2030)	Error Code: 3819. Check constraint 'chk_published_year' is violated.	0.000 sec
---	----	----------	---	--	-----------

Test 4: Duplicate email (UNIQUE violation)

```
INSERT INTO members (full_name, email)
VALUES 'G. Hemanth Venkatesh', 'hemanthvenkatesh5554@gmail.com'
```

Output: -

✖	19	10:57:15	INSERT INTO members (full_name, email) VALUES ('Adama Samarth', 'adamasamarth@gmail.com')	Error Code: 1062. Duplicate entry 'adamasamarth@gmail.com' for key 'members.email'	0.031 sec
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